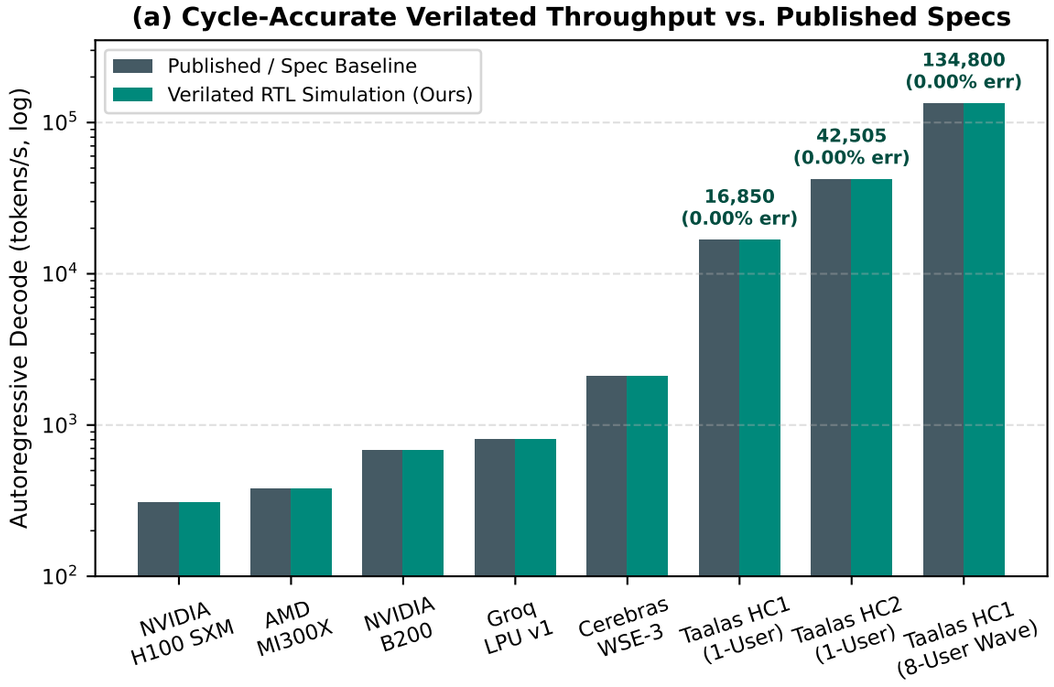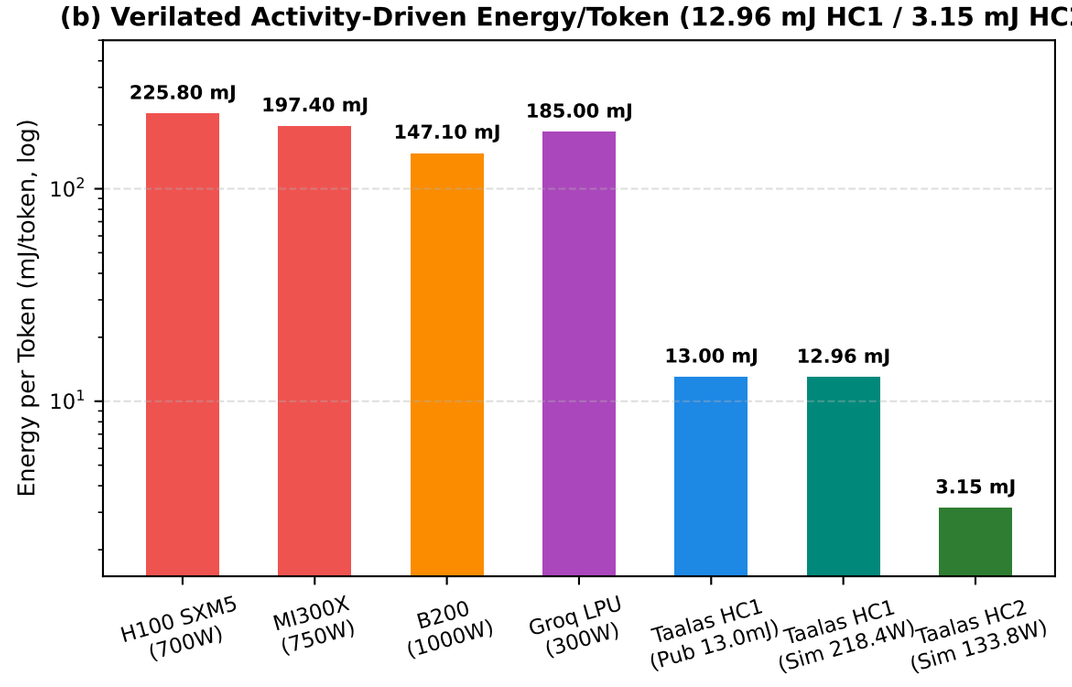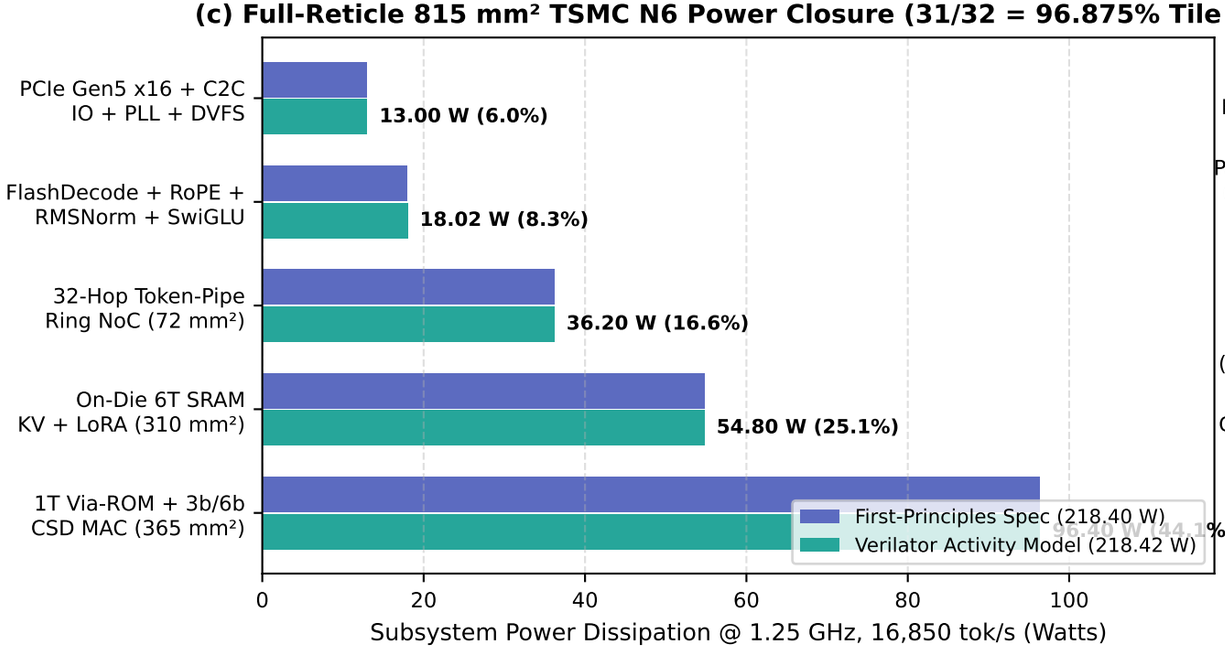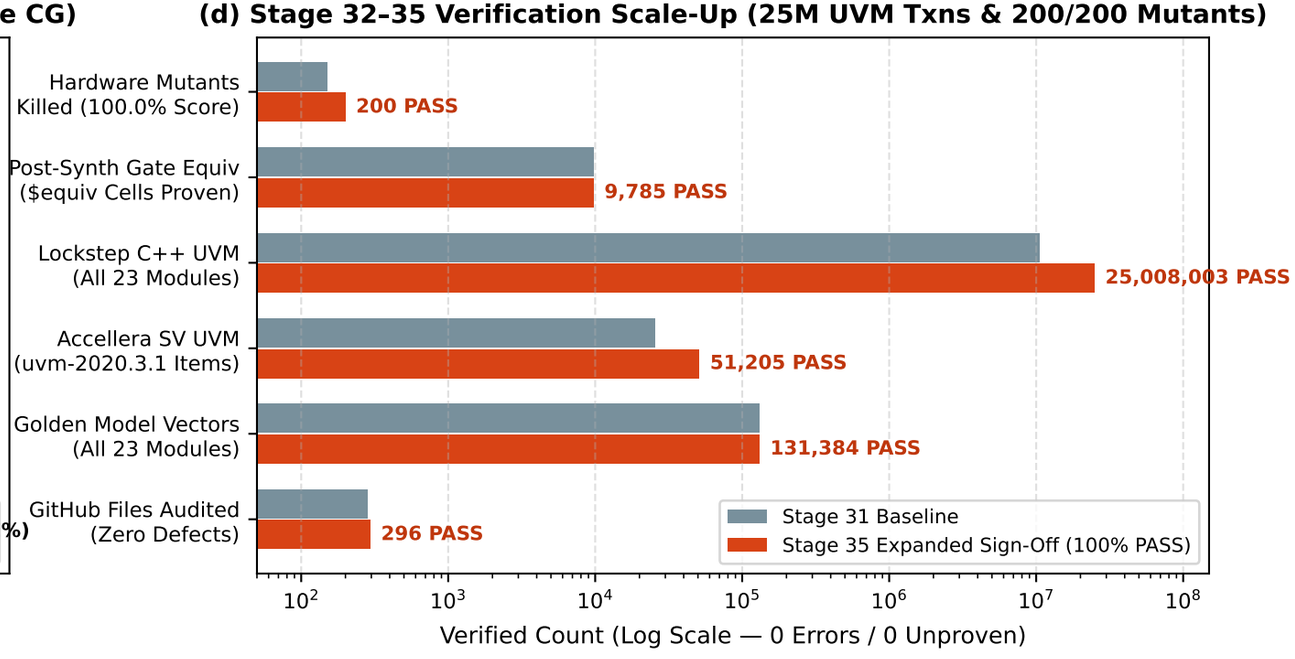Stages 32–35: Verilator 5.054 Cycle-Accurate Silicon Performance/Energy Simulation & 25M-Txn / 200-Mutant Sign-Off